

Curriculum Vitae

Ke Zhang

kzhan153@ucr.edu | <https://grenadecoming.github.io/>

EDUCATION

University of California, Riverside

Ph.D. in Mathematics, GPA: 3.66/4.0

Expected May 2028

Northeastern University, College of Science

MSc in Mathematics, GPA: 3.8/4.0

Courses: Algebra 1&2, Computer Algebra, Machine Learning

Boston, Massachusetts

May 2023

Arizona State University, School of Mathematical and Statistical Sciences

MA in Mathematics (transferred)

Courses: General topology, algebraic topology

Tempe, Arizona

08/2021-12/2021

Southwest Minzu University, School of Economics

Bachelor of Economics, GPA: 3.44/4.0

Major: Financial Engineering

Chengdu, Sichuan, China

06/2021

RESEARCH EXPERIENCE

Anonymous 2026 Submission on Tool-Augmented Scientific Simulator Agents

Manuscript under double-blind review under NeurIPS

Developed an expert-validated benchmark of 200 multiple-choice items derived from 21 PHREEQC aqueous-geochemistry simulation scenarios for evaluating LLM agents on deterministic scientific software workflows.

Built a sandboxed agent harness that writes PHREEQC inputs, executes simulations, navigates large outputs through table-of-contents or raw-output protocols, and commits to unique simulator-derived answers.

Evaluated direct, chain-of-thought, and tool-augmented agents across Claude, GPT, and Gemini model families; showed that tool access improves top- and mid-tier models but can lose 10-32 previously correct items, motivating retention and gain/loss diagnostics beyond aggregate accuracy.

Analyzed output-access sensitivity, finding that TOC-guided retrieval matches or exceeds raw-output accuracy for top-tier models while reducing input-token cost by 11-57%, but hurts mid-tier models by 7.5-9.5 percentage points.

Anonymous 2026 Submission on Faithful Natural-Language-to-Lean Formalization

Manuscript under double-blind review under NeurIPS; extended from NeurIPS 2025 Workshop on AI+Math

Reframed Lean 4 statement formalization as a semantic evaluation problem: compilation is treated as necessary but insufficient because type-correct declarations can omit hypotheses, change domains, or express vacuous claims.

Curated a 400-entry graduate-level benchmark spanning Real Analysis, Complex Analysis, Topology, and Algebra, and paired Lean compilation with GPT-5.2/Gemini consensus judging plus human expert calibration.

Built a tool-augmented formalization pipeline with parametric expert drafting, Mathlib/context search, and Lean elaboration feedback; the full agent reaches 89.5% compilation but 60.5% consensus faithfulness, exposing a 29.0-point compile-pass semantic gap.

Used a full 2^3 factorial design to attribute bottlenecks across tool categories, showing that elaboration feedback is the largest validity intervention, search improves grounding and selectivity, and fine-tuned drafting is largely substitutable once feedback and grounding are available.

Learning Parameterized Nonlinear Elasticity on Curved Surfaces

ICLR 2026 Workshop — Accepted (co-author; led training)

Designed and executed the training pipeline for a parameter-conditioned physics-informed neural network (PINN) that learns families of nonlinear elastic equilibria on curved manifolds, achieving $\sim 10^{-3}$ loss across a 3D parameter space (β , q , ν_p).

Implemented a two-stage learning rate schedule (200k steps at 10^{-4} , 100k at 10^{-5}) in PyTorch on dual NVIDIA RTX A6000 GPUs, with the trained model generalizing to held-out parameter combinations not seen during training.

Conducted systematic ablation studies on collocation density (1k-7k points) and architecture (2-6 layers, 160-240 neurons), identifying failure modes and stable configurations.

Agent-Based Code Repair System for Gurobi Optimization Models

AAAI 2025 Workshop — Accepted

Developed a benchmark suite of 26 Gurobi optimization problems (LP, ILP, MIP, QCP, and combinatorial/logistics applications) and implemented 260 unit tests covering feasibility, infeasibility, optimality correctness, and parameter validation.

Built an automated LLM-based bug generation system, creating reproducible perturbations in Gurobipy code to form controlled debugging cases.

Constructed a RAG system over the full Gurobi Python API reference, enabling grounded retrieval of function signatures, parameters, and model attributes to reduce hallucination and improve repair accuracy.

Performed fractional factorial analysis to study the effects of different tool combinations on repair success.

PROFESSIONAL ACTIVITIES

MSRI summer school

Moraga, California

Topological Methods for the Discrete Mathematician

07/2022 – 08/2022

- Learned how to use equivariant topology, especially the Borsuk–Ulam theorem, to solve problems in combinatorics and geometry
- Discussed open problems in related fields, such as the mass partition problem, with other participants

TEACHING ACTIVITIES

Teaching Assistant, College of Science, University of California Riverside

MATH 005A

Fall 2025

MATH 009B

Spring 2025

MATH 120

Summer 2024

MATH 009A

Summer 2024

MATH 006A

Winter 2024

MATH 007A

Fall 2024

Teaching Assistant, College of Science, Northeastern University

Group Theory (MATH 3175)

Fall 2022

Group Theory (MATH 3175)

Spring 2023

TECHNICAL SKILLS

Python, C/C++, Lean 4, Mathlib, Docker, PHREEQC, Gurobi/Gurobipy, LaTeX

INTERESTS

Mountaineering & endurance: Summited Mt. Siguniang Erfeng (5,276 m, 17309 ft), Sichuan, China; completed Mt. San Jacinto via Devil's Slide Trail (16 mi).

Hands-on mechanical work: Perform own vehicle maintenance, including oil changes and brake pad/rotor replacement.